

If we ignore v_{om} , then the differential output op-amp behaves just like a single-ended op-amp. For linear applications, the op-amp is used with a feedback network, the inputs are related (assuming the open-loop gain, A_{OL} is large) by

$$v_p \approx v_m \quad (25.11)$$

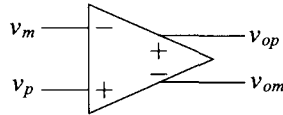


Figure 25.12 Differential output op-amp.

Common-Mode Feedback (CMFB)

Normally, the average of the op-amp outputs is called the common-mode output voltage. The two op-amp outputs swing around the common-mode voltage. If the largest op-amp output voltage is V_{DD} and the minimum output voltage is ground, then the common-mode voltage is

$$V_{CM} = \frac{V_{DD}}{2} \quad (25.12)$$

Figure 25.13a shows a simple differential output op-amp gain configuration (inverting or noninverting, depending on which output is used as the positive output). Because the input voltages to the circuit are equal, the output voltages of the op-amp should remain at V_{CM} . Due to the op-amp's high gain and the negative feedback, we always have $v_p \approx v_m$ (Eq. [25.11]). However, if $v_{op} = v_{om} = V_{DD}$ or $v_{op} = v_{om} = \text{ground}$ or $v_{op} = v_{om} = \text{anything}$, then $v_p = v_m$ and Eq. (25.11) is satisfied. *This is a problem* and the reason why we need a CMFB circuit. In Fig. 25.13b, we add a CMFB circuit to monitor the outputs of the op-amp and make adjustments in the op-amp (how these adjustments are made is discussed in the next chapter) to keep the two outputs balanced around V_{CM} ($V_{DD}/2$).

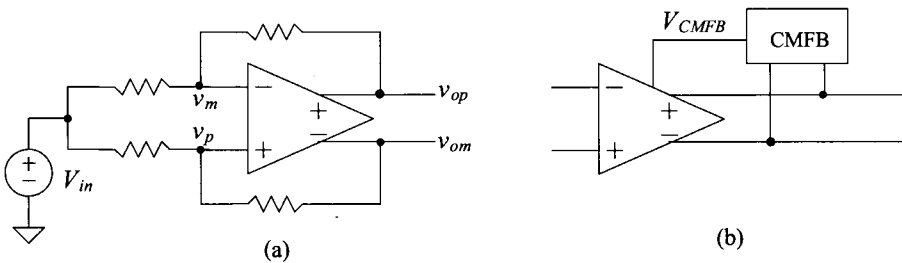


Figure 25.13 (a) Simple gain configuration using differential output op-amp and (b) the use of a common-mode feedback circuit to adjust the common-mode output voltage.

Coupled Noise Rejection

Differential topologies are required to minimize the effects of charge injection and capacitive feedthrough from switches. They also are used to help reduce the effects of coupled noise. Consider the cascade of differential output op-amps, with no feedback network shown, seen in Fig. 25.14. The stray capacitance between the interconnecting metal lines (the metal lines that carry the signals between op-amps) and the substrate or any other noise source are shown. If the metal lines are run close to one another, then the noise voltage will couple even amounts (ideally) of noise into each signal wire. Since the diff-amp, on the input of the op-amp, rejects common signals (signals that are present on both inputs), the coupled noise is not passed to the next op-amp in the string. Variations on the power supply rails are rejected as well. If the differential op-amp is symmetric, then changes on the power supply couple evenly into both outputs, having little effect on the difference, or desired signal, coming out of the op-amp. For these reasons, that is, good coupled noise rejection and PSRR, the differential op-amp is a necessity in any dynamic analog integrated circuit.

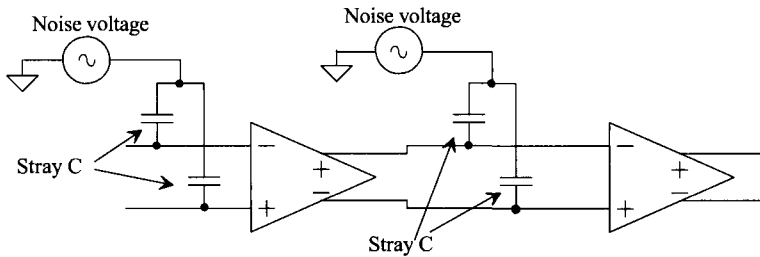


Figure 25.14 Differential output op-amps showing parasitic capacitance and noise.

Other Benefits of Fully-Differential Op-Amps

Another benefit of fully-differential circuits is a doubling in output voltage swing. For example, if V_{DD} is 1 V, then the output of a single-ended op-amp can swing from 1 V to ground. In a fully-differential topology, both v_{op} and v_{om} can swing from 0 to 1. Using Eq. (25.9), we can write

$$v_{out,max} = v_{op} - v_{om} = 1 - 0 = 1 \text{ V and } v_{out,min} = v_{op} - v_{om} = 0 - 1 = -1 \text{ V} \quad (25.13)$$

The output swing is $2V_{DD}$ when using fully-differential topologies.

A further benefit of fully-differential topologies is the fact that the input common-mode voltage of the op-amp remains at V_{CM} (and so the op-amp's first stage diff-amp's common-mode voltage range requirements are easy to meet).

25.2.1 A Fully-Differential Sample-and-Hold

Figure 25.15 shows a fully-differential sample-and-hold circuit and the associated clock waveforms that eliminate clock feedthrough and charge injection to a first order. The switches in this figure are closed when their controlling clock signals are high. The basic operation can be understood by considering the state of the circuit at t_0 . At this time, the

input signals charge the sampling capacitors. The bottom plates of the capacitors (poly1) are tied directly to the input signals, for reasons that will be explained below. The op-amp is operating in a unity-follower configuration in which both inputs of the op-amp are held at V_{CM} . At this particular instance in time, prior to t_1 , the amplifier is said to be operating in the sample mode of operation.

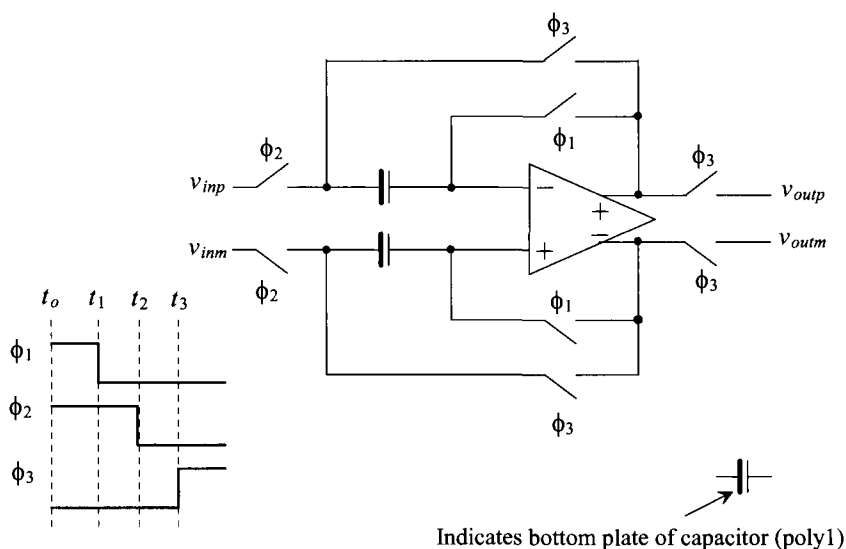


Figure 25.15 Sample-and-hold using differential topology.

At t_1 , the ϕ_1 switches turn off. The resulting charge injection and clock feedthrough appear as a common-mode signal on the inputs of the op-amp and are ideally rejected. Since the top plates of the hold capacitors (the inputs to the op-amp) are always at V_{CM} , at this point in time the charge injection and clock feedthrough are independent of the input signals. This produces an increase in the dynamic range of the sample-and-hold (the minimum measurable input signal decreases). The voltage on the inputs of the op-amp (the top plate of the capacitor) between t_1 and t_2 is $V_{OFF1} + V_{CM}$, a constant voltage. Note that the op-amp is operating open loop at this time so the time between t_1 and t_3 should be short.

At t_2 the ϕ_2 switches turn off. At this point in time, the voltages on the bottom plates of the sampling (or hold) capacitors (poly1) are v_{inp} and v_{inm} . The voltages on the top plates of the capacitors (connected to the op-amp) are $V_{OFF1} + V_{OFF2} + V_{CM}$ (assuming that the storage capacitors are much larger than the input capacitance of the op-amp). The term V_{OFF2} is ideally a constant that results from the charge injection and capacitive feedthrough from the ϕ_2 switches turning off. The time between t_1 and t_2 should be short compared to variations in the input signals.

At time t_3 the ϕ_3 switches turn on and the op-amp behaves like a voltage follower; the circuit is said to be in the hold mode of operation. The charge injection and clock feedthrough resulting from the ϕ_3 switches turning on causes the top plate of the capacitor to become $V_{OFF1} + V_{OFF2} + V_{OFF3} + V_{CM}$, again assuming that the storage capacitors are much larger than the input capacitance of the op-amp. The outputs of the sample-and-hold are v_{inp} and v_{inn} , assuming infinite op-amp gain since these offsets appear as a common-mode voltage on the input of the op-amp. Note that the terms V_{OFF2} and V_{OFF3} are dependent on the input signals.

Connecting the Inputs to the Bottom (Poly1) Plate

The reason for connecting the input signals to the bottom plate of the capacitor can be explained with the help of Fig. 25.16. This figure is a simplified, single-ended version of Fig. 25.15 where the capacitance, C_p , is the parasitic capacitance from the bottom plate to the substrate. With regard to Fig. 25.16a, coupled noise from the substrate sees either the input voltage from the op-amp driving the sample-and-hold (in the sample mode) or the output voltage of the op-amp used in the sample-and-hold itself (in the hold mode). Since the op-amps in either mode directly set this voltage, the substrate noise has, ideally, little effect on the circuit's operation.

In Fig. 25.16b, coupled substrate noise feeds directly into the input of the op-amp and can thus drastically affect the output of the sample-and-hold. Another more subtle problem occurs in the circuit of Fig. 25.16b. When the circuit makes the transition to the hold mode at t_3 , the output of the op-amp should quickly change to the voltage sampled on the input capacitors. The time it takes the output of the op-amp to change and settle to this final voltage is called the *settling time*. The parasitic capacitance on the input of the op-amp in (b) reduces the feedback factor from unity to $C_H / (C_p + C_H)$. This slows the settling time and causes a gain error in the circuit's transfer function. For these reasons, the parasitic capacitance on the top plate of the capacitor should be small. Nothing should be laid out over or in near proximity of poly2.

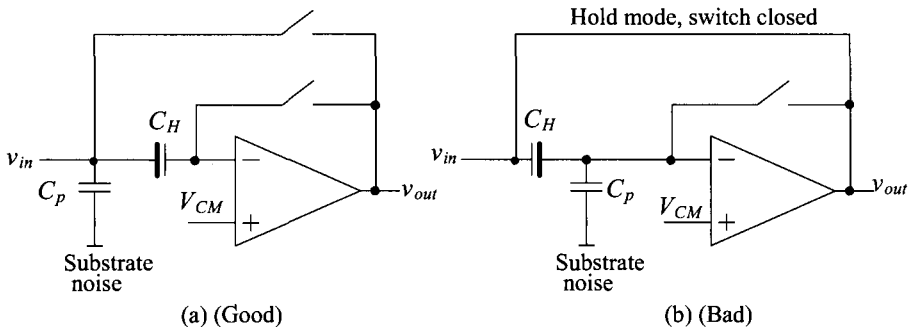


Figure 25.16 Explanation for connecting the bottom plate of the capacitor to the input.

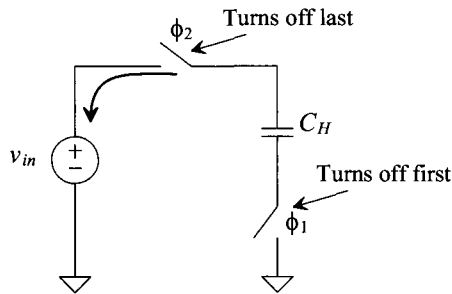


Figure 25.17 Bottom plate sampling.

Bottom Plate Sampling

Turning the ϕ_1 controlled switches off in Fig. 25.15 slightly before the ϕ_2 controlled switches is sometimes called *bottom plate sampling*. Figure 25.17 illustrates why. When the ϕ_1 switches turn off, the charge injected into C_H is a constant independent of the input signal amplitude. The resulting offset voltage across C_H is then constant. When the ϕ_2 switches turn off, the resulting charge injection then takes the path of least resistance, that is, into the input source v_{in} . The voltage across C_H is then, ideally, independent of the input signal voltage.

SPICE Simulation

Let's simulate the operation of the sample-and-hold in Fig. 25.15. To begin, let's use voltage-controlled voltage sources and a DC voltage to model the op-amp (again, we'll discuss differential output op-amps in the next chapter). The ideal fully-differential output op-amp SPICE model is seen in Fig. 25.18. The SPICE netlist statements used to model this circuit (we'll assume an open-loop gain of 10^6 and VDD of 1 V) are

```
E1 vop vcm vp vm 1e6
E2 vcm vom vp vm 1e6
Vcm vcm 0 DC 500m
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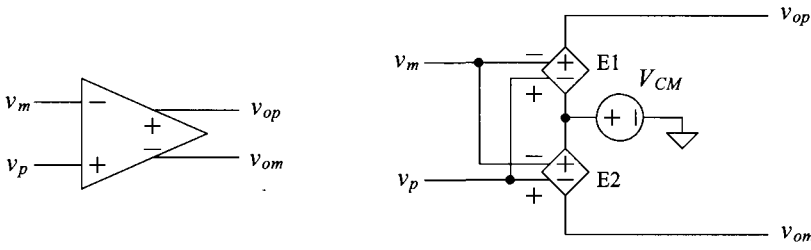


Figure 25.18 SPICE modeling a differential input/output op-amp with common-mode voltage.

For the switches we'll use NMOS devices as seen in the simulation schematic of Fig. 25.19. For the input signals, we've used a 5 MHz sinewave signal with a peak amplitude of 200 mV. Note that each of the inputs is referenced to the common-mode voltage, V_{CM} (which is 500 mV here).

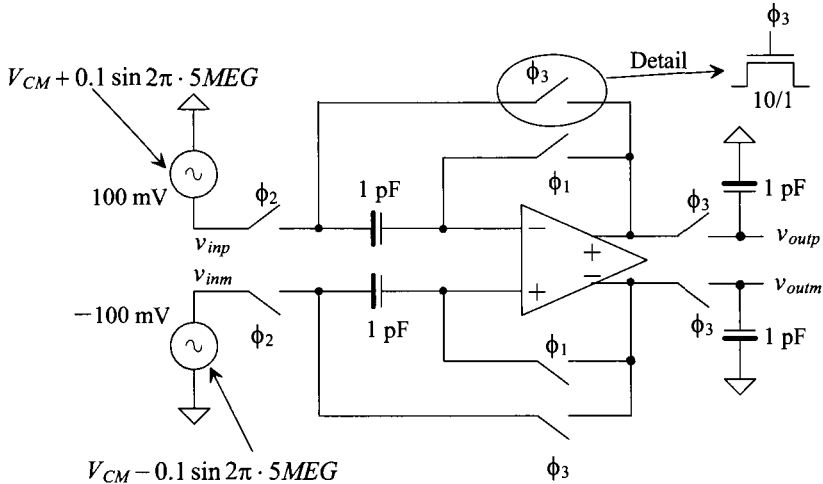


Figure 25.19 Simulating the operation of the fully differential sample-and-hold.

The simulation results are seen in Fig. 25.20. Let's first focus on the clock signals. Notice how ϕ_3 isn't high at the same time as ϕ_1 or ϕ_2 . If all switches were "on" at the same time, the charge stored (or held) on the 1 pF capacitors at the output of the circuit would change (remember when the ϕ_1 switches close the op-amp is in the follower configuration with the inputs and outputs pulled to V_{CM}). We make sure to disconnect these output capacitors before putting the circuit into the sample mode. The output of the sample-and-hold seen in Fig. 25.20 doesn't exactly match the input signal. There is a one-clock cycle delay and we can see the effects of the finite clock frequency (finite sample points). Note, these topics are discussed in detail in the book entitled *CMOS Mixed-Signal Circuit Design*.

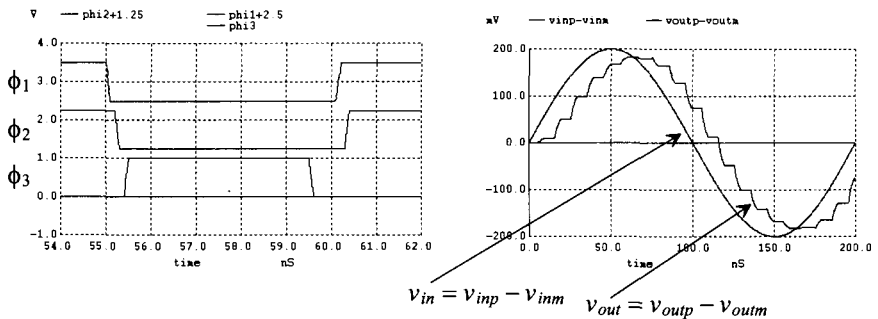


Figure 25.20 Simulating the operation of the sample-and-hold in Fig. 25.19.

25.3 Switched-Capacitor Circuits

Consider the circuit shown in Fig. 25.21a. This dynamic circuit, named a *switched-capacitor resistor*, is useful in simulating a large value resistor, generally $>1\text{ M}\Omega$. The clock signals ϕ_1 and ϕ_2 form two phases of a nonoverlapping clock signal with frequency f_{clk} and period T as seen in the figure. Let's begin by considering the case when S1 is closed. When ϕ_1 is high, the capacitor C is charged to v_1 . The charge, q_1 , stored on the capacitor during this interval, Fig. 25.21c, is

$$q_1 = Cv_1 \quad (25.14)$$

while if S2 is closed, the charge stored on the capacitor is

$$q_2 = Cv_2 \quad (25.15)$$

If v_1 and v_2 are not equal, keeping in mind that S1 and S2 cannot be closed at the same time due to the nonoverlapping clock signals, then a charge equal to the difference between q_1 and q_2 is transferred between v_1 and v_2 during each interval T . The difference in the charge is given by

$$q_1 - q_2 = C(v_1 - v_2) \quad (25.16)$$

If v_1 and v_2 vary slowly compared to f_{clk} , then the average current transferred in an interval T is given by

$$I_{avg} = \frac{C(v_1 - v_2)}{T} = \frac{v_1 - v_2}{R_{sc}} \quad (25.17)$$

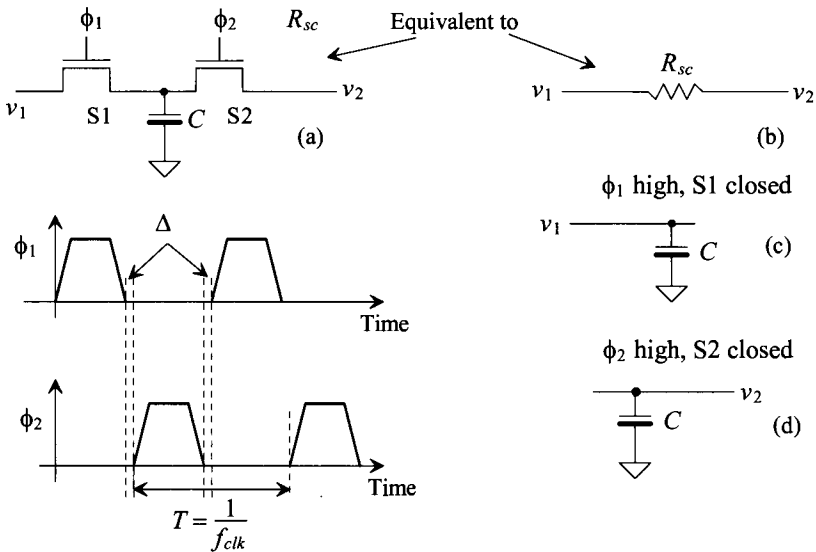


Figure 25.21 Switched-capacitor resistor (a) and associated waveforms and (b, c, d) the equivalent circuits.

The resistance of the switched-capacitor circuit is given by

$$R_{sc} = \frac{T}{C} = \frac{1}{C \cdot f_{clk}} \quad (25.18)$$

In general, the signals v_1 and v_2 should be bandlimited to a frequency at least ten times less than f_{clk} (more on this later). Note that we derived a similar result back in Ch. 17, see Eq. (17.29).

Example 25.1

Using switched-capacitor techniques, implement the circuit shown in Fig. 25.22a so that the product of RC is 1 ms, that is, the 3-dB frequency of $|v_{out}/v_{in}|$ is 159 Hz.

The switched-capacitor implementation of this circuit is shown in Fig. 25.22b. The product of RC may now be written in terms of Eq. (25.18) as

$$RC_2 = \frac{C_2}{C_1} \cdot \frac{1}{f_{clk}} \quad (25.19)$$

This result is important! The product RC_2 is determined by f_{clk} , which may be an accurate frequency derived from a crystal oscillator and the ratio of C_2 to C_1 , which will be within 1% on a chip. This means that even if the values of the capacitors change by 20% from wafer to wafer, the ratio of the capacitors relative to one another, on the same wafer, will remain constant within 1%.

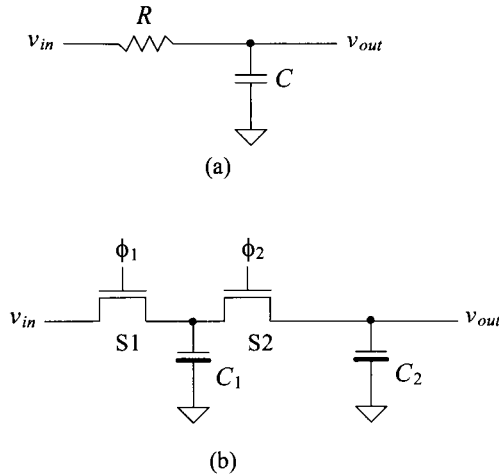


Figure 25.22 (a) Circuit used in Ex. 27.1 and (b) its implementation using a SC resistor.

It is also desirable to keep C_1 larger than the associated parasitics present in the circuit (e.g., depletion capacitances of the source/drain implants and the stray capacitances to substrate). For the present example, we will set C_1 to 1 pF. The selection of f_{clk} is usually determined by what is available. For the present design, a value of 100 kHz will be used. This selection assumes that the energy present in v_{in} at frequencies above 10 kHz is negligible. The value of C_2 is determined solving Eq. (25.19) and is 100 pF. Note that the value of the switched-capacitor

resistor is 10 M Ω . Implementing this resistor in a CMOS process using n-well with a sheet resistance of 1,000 ohms/square would require 10,000 squares! The resulting delay through the n-well resistor, because of the capacitance to substrate, may cause a significant phase error in the transfer function. ■

25.3.1 Switched-Capacitor Integrator

Because the switched-capacitor resistor of Fig. 25.21a is sensitive to parasitic capacitances, it finds little use, by itself, in analog switched-capacitor circuits. Consider the circuit of Fig. 25.23a. This circuit, a switched-capacitor integrator, is the heart of the circuits we will be discussing in the remainder of the section. The portion of the circuit consisting of switches S1 through S4 and C_I forms a switched-capacitor resistor with a value given by

$$R_{sc} = \frac{1}{C_I f_{clk}} \quad (25.20)$$

The equivalent continuous time circuit for the switched-capacitor integrator is shown in Fig. 25.23b. Notice that v_{in} is now negative. It may be helpful in the following discussion to remember that the combination of switches and C_I of the switched-capacitor integrator can be thought of as a simple resistor. The transfer function of the switched-capacitor integrator is given by

$$\frac{v_{out}}{v_{in}} = \frac{1/j\omega C_F}{R_{sc}} = \frac{1}{j\omega \left(\frac{C_F}{C_I} \cdot \frac{1}{f_{clk}} \right)} \quad (25.21)$$

Again, the ratio of capacitors is present, allowing the designer to precisely set the gain of the amplifier and the integration time constant.

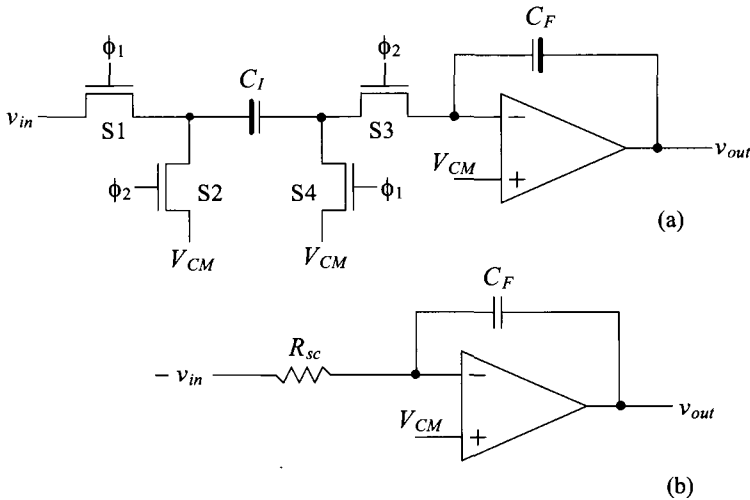


Figure 25.23 (a) A stray insensitive switched-capacitor integrator (noninverting) and (b) the equivalent continuous time circuit.

Parasitic Insensitive

The switched-capacitor integrator in Fig. 25.23 is not sensitive to parasitic or stray capacitances. This can be understood with the use of Fig. 25.24. To begin, if we realize that C_{p2} (the parasitic capacitance on the right side of C_I) is always connected to V_{CM} either through S4 or through the connection to the inverting input of the op-amp, then C_{p2} doesn't see a change in the charge stored on it. Next, the capacitance C_{p1} is charged to v_{in} when S1 is closed and then charged to V_{CM} when S2 closes. Since none of the charge stored on C_{p1} when S1 is closed is transferred to C_I , it does not affect the integrating function. A practical minimum for C_I is 100 fF set by kT/C noise (see Table 8.1).

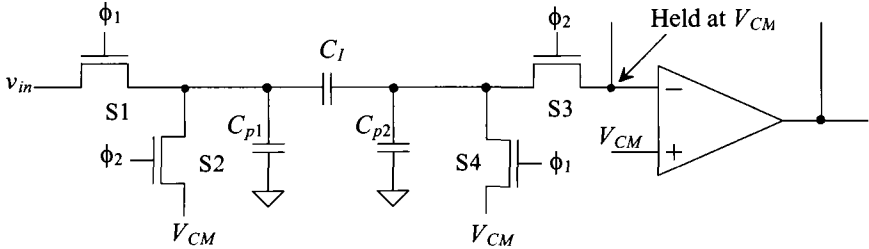


Figure 25.24 Parasitic capacitances associated with a switched-capacitor resistor.

Other Integrator Configurations

An inverting integrator configuration can be formed by simply swapping the clock signals used with S1 and S2 in Fig. 25.23 (the noninverting configuration). The gate of S1 is connected, for the inverting configuration, to ϕ_2 while the gate of S2 is connected to ϕ_1 . The gain of this configuration is given by

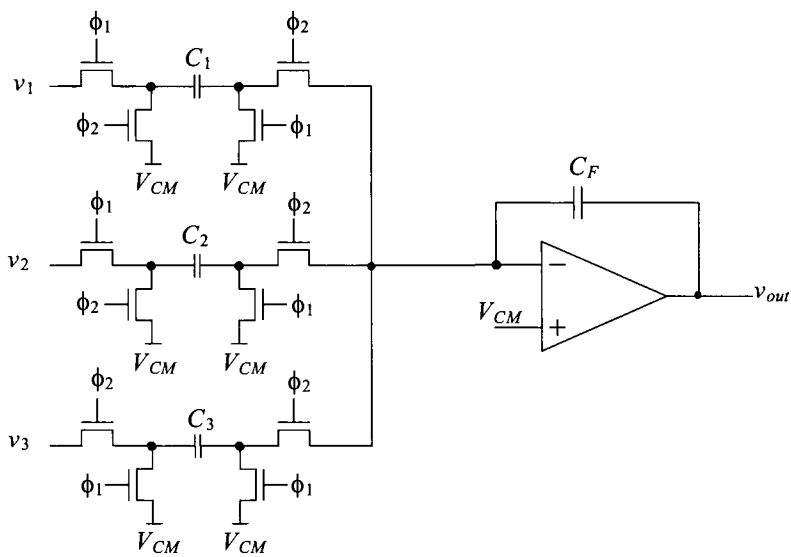
$$\frac{v_{out}}{v_{in}} = -\frac{1}{j\omega \left(\frac{C_F}{C_I} \cdot \frac{1}{f_{clk}} \right)} \quad (25.22)$$

An example of a switched-capacitor integrator circuit that combines input signals is shown in Fig. 25.25a. Remembering that each switched-capacitor section can be thought of as a resistor, we note that the relationship between the inputs and output is

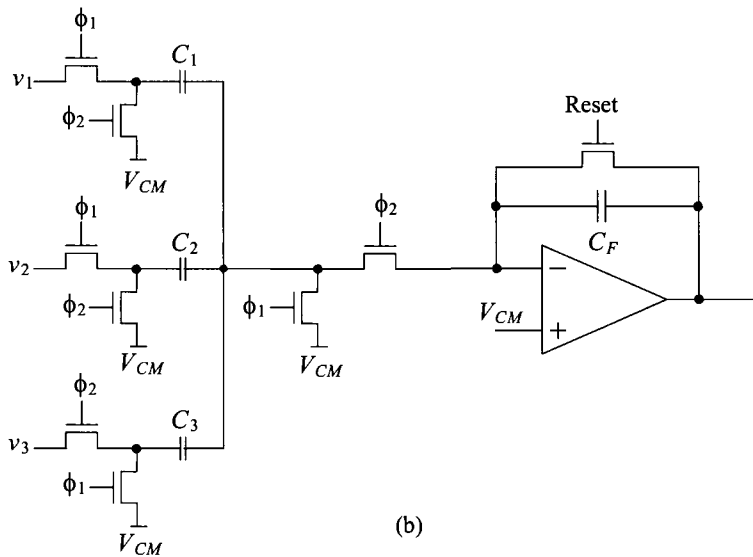
$$v_{out} = \frac{v_1}{j\omega \left(\frac{C_F}{C_1} \frac{1}{f_{clk}} \right)} + \frac{v_2}{j\omega \left(\frac{C_F}{C_2} \frac{1}{f_{clk}} \right)} - \frac{v_3}{j\omega \left(\frac{C_F}{C_3} \frac{1}{f_{clk}} \right)} \quad (25.23)$$

Figure 25.25b shows how redundant switches can be combined to reduce the number of devices used. Used alone, the basic integrator has the practical problem of integrating not only the input signal but also the offset voltage of the op-amp. In many applications, a reset switch or resistor is placed across the feedback capacitor (Fig. 25.25b). An example of a lossy integrator circuit useful in first-order filter design is shown in Fig. 25.26. The transfer function of this circuit is given by

$$\frac{v_{out}}{v_{in}} = \frac{R_4}{R_3} \left(\frac{1 + j\omega R_3 C_1}{1 + j\omega R_4 C_2} \right) = \frac{C_3}{C_4} \left(\frac{1 + j\omega \left(\frac{C_1}{C_3} \cdot \frac{1}{f_{clk}} \right)}{1 + j\omega \left(\frac{C_2}{C_4} \cdot \frac{1}{f_{clk}} \right)} \right) \quad (25.24)$$



(a)



(b)

Figure 25.25 (a) Switched-capacitor implementation of a summing integrator and (b) practical implementation of the circuit combining switches and adding reset.